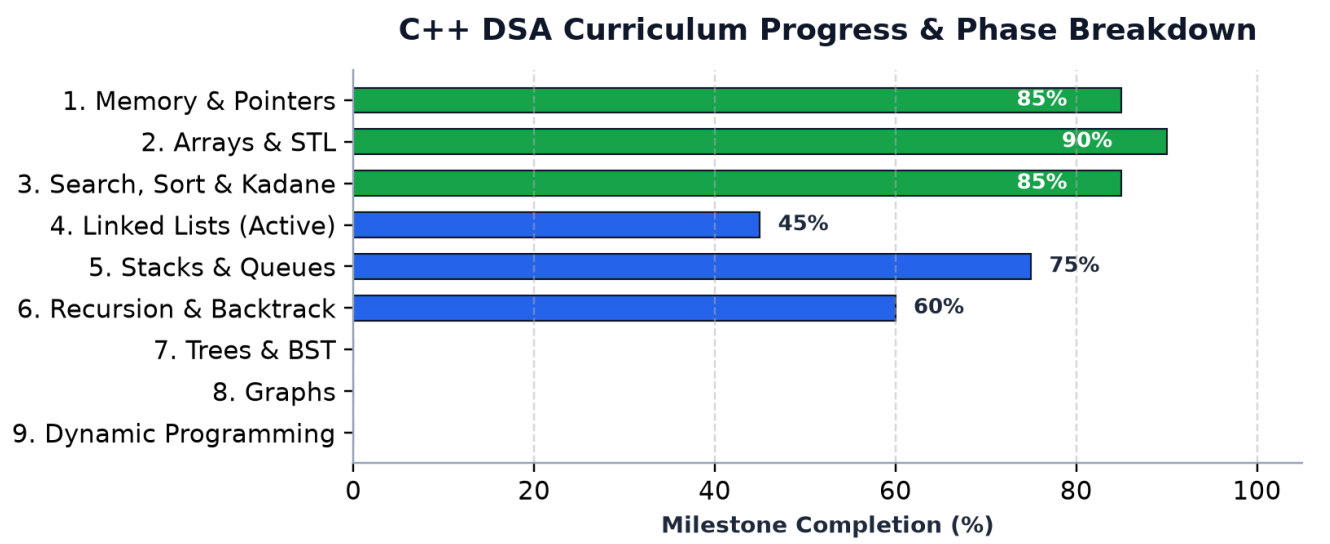


C++ DSA Master Learning Tracker

Official Persistent Study Record — Algorithms, Hardware Architecture & LeetCode Mastery

Student: Amit	Current Focus: Phase 4 (Linked Lists)	Language: C++ (C++17 / C++20)
Target: LeetCode Medium/Hard & Interviews	Active Exercise: PRACTICE.cpp	Track Status: In Active Progress



Curriculum Roadmap & Milestone Status

Phase	DSA Domain	Key Topics Covered	Status
Phase 1	Memory & Pointers	Stack vs Heap, pointers (*, &), double pointers, pointer arithmetic	85% Mastered
Phase 2	Linear DS & STL	Arrays, 2D Arrays, Vectors, Strings, Pair, Set, Map, Deque, List	90% Mastered
Phase 3	Algorithms & Sorting	Binary Search, Merge Sort, Kadane's (\$O(N)\$), Moore's Voting	85% Mastered
Phase 4	Linked Lists	Node class, chaining, insertion at head/tail, delete, search	45% Active
Phase 5	Stacks & Queues	LIFO/FIFO, Stock Span, Next Greater Element, Priority Queue (Min/Max Heap)	75% Mastered
Phase 6	Recursion	Call stack, recursion trees, factorial, fibonacci, backtracking	60% Mastered
Phase 7	Trees & BST	Binary trees, traversals, height/diameter, BST properties	Upcoming
Phase 8	Graphs	BFS, DFS, Cycle detection, Dijkstra, Shortest path	Upcoming
Phase 9	Dynamic Prog.	Memoization, Tabulation, 1D/2D DP, Knapsack, LCS	Upcoming

Study Memory & Pedagogical Log

Session Focus (Phase 4): Linked Lists & Pointer Architecture.

Repository Optimization: Cleaned entire workspace, standardized all directory names to snake_case, moved loose files to linked_list/, established two_pointers/ algorithm directory, and eliminated binary executables from version control.

Next Immediate Steps: Dynamic Linked List traversal using while loops, pointer safety checks (nullptr guards), Head & Tail pointer updates, and Socratic calibration exercises in PRACTICE.cpp.